



Global Landslide Catalog

1988 – 2017 · 11,033 recorded events · NASA GLC

Overview

Causes & Severity

Time & Geography

DATASET OVERVIEW

Source: NASA Global Landslide Catalog (GLC)

URL: <https://catalog.data.gov/dataset/global-landslide-catalog-export>

Time Period: 1988 – 2017 (30 years)

Total Records: 11,033 landslide events

Geographic Coverage: 142 countries

Original Columns: 31

Final Columns (after cleaning): 27

Key Fields:

- Event Date, Country, Admin Division, Location
- Landslide Category (Landslide, Mudslide, Rock Fall, etc.)
- Landslide Trigger (Downpour, Earthquake, Monsoon, etc.)
- Fatality Count, Injury Count
- Latitude, Longitude (for mapping)

Challenges Encountered

1. Mixed Date Formats

Event dates in MM/DD/YYYY and MM-DD-YYYY
Solution: Custom M formula with try/otherwise

2. Character Encoding (Mojibake)

Special characters displayed incorrectly
Solution: Set file origin to UTF-8

3. Inconsistent Text Casing

snake_case values in categorical fields
Solution: Capitalize + replace underscores

4. Null Values Strategy

Many columns had null entries
Solution: null→0 for counts,
null→"Unknown" for text

5. Derived Fields Added

Year Month Quarter Day of Week



Click to Check The Published File



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Methodology

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Year

All

Severity Bucket

All

Landslide Trigger

- ☐ Construction
- ☐ Continuous Rain
- ☐ Dam Embankment Collap.
- ☐ Downpour
- ☐ Earthquake
- ☐ Flooding
- ☐ Freeze Thaw
- ☐ Leaking Pipe
- ☐ Mining
- ☐ Monsoon
- ☐ No Apparent Trigger

11,033

Total Events

31,061

Total Fatalities

4,029

Total Injuries

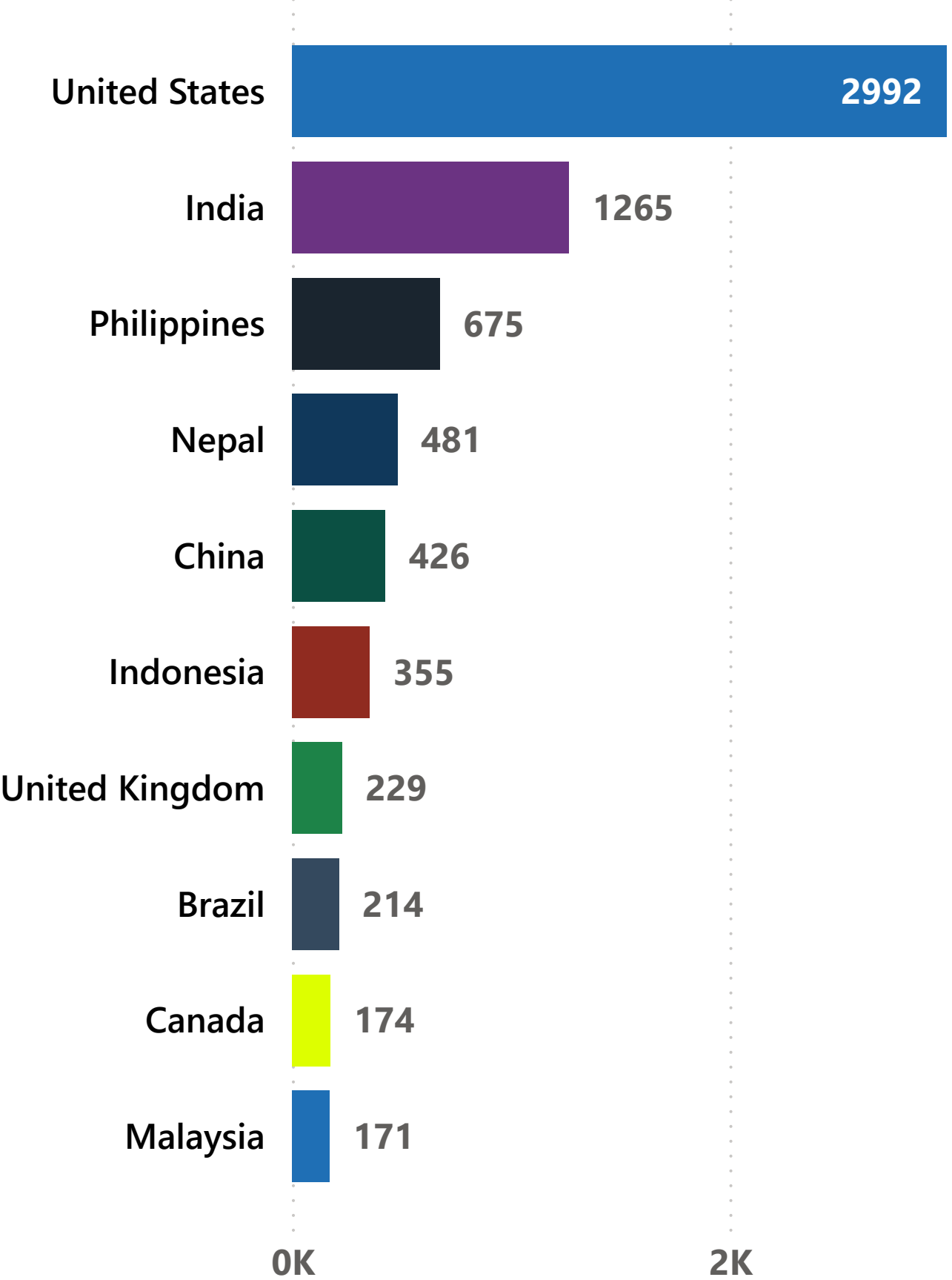
142

Countries Affected

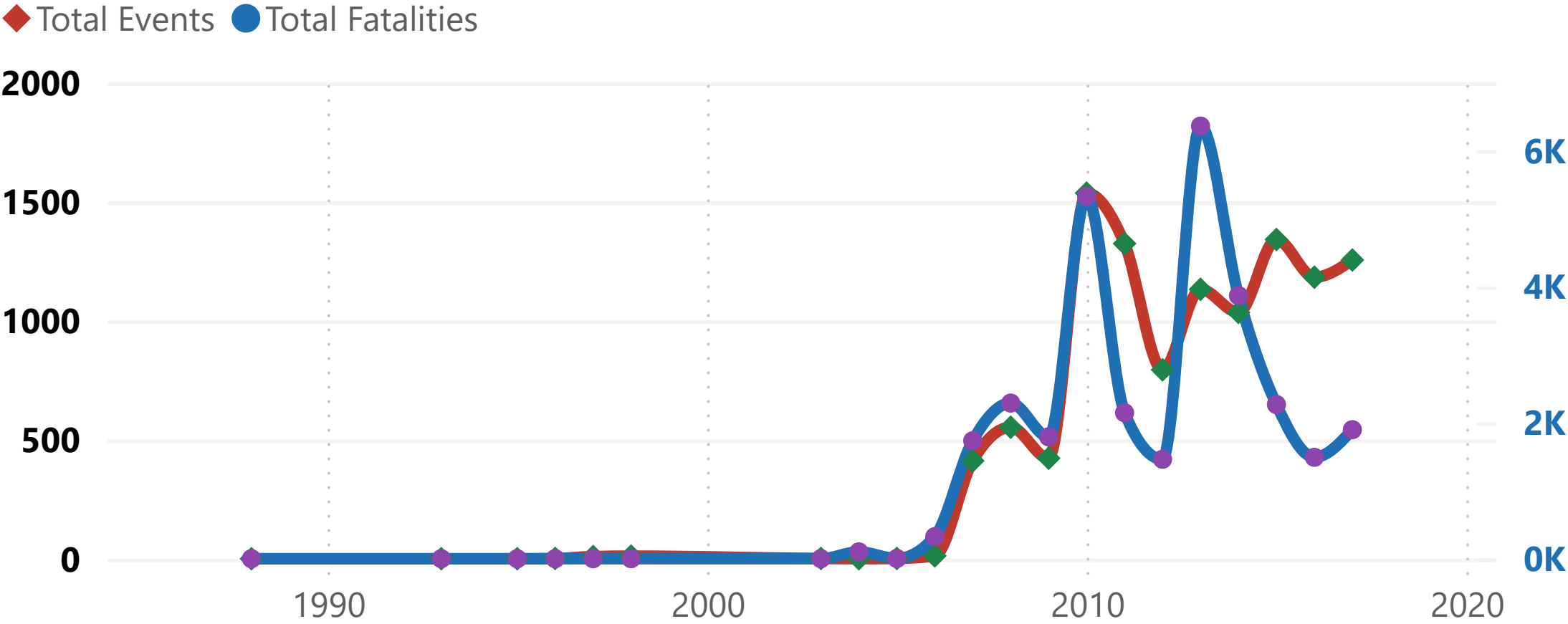
30

Years Covered

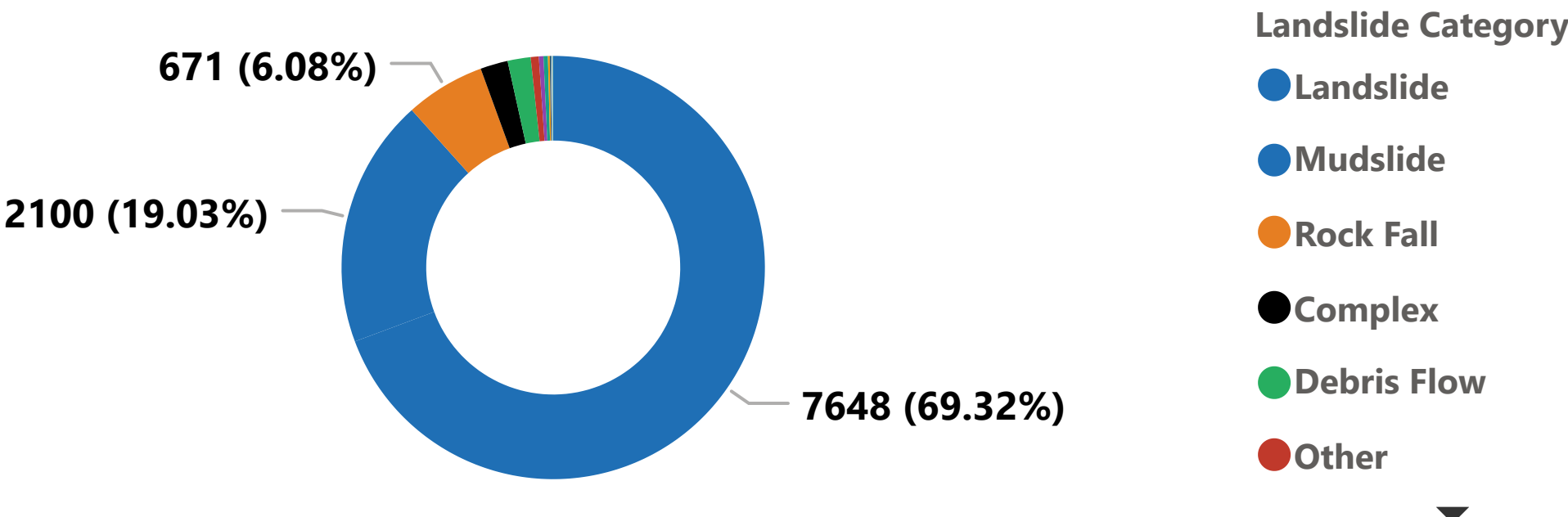
Total Events by Country Name



Events & Fatalities Over Time



Landslide Category





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Year

All

Country Name

All

Landslide Category

- ☐ Complex
- ☐ Creep
- ☐ Debris Flow
- ☐ Earth Flow
- ☐ Lahar
- ☐ Landslide
- ☐ Mudslide
- ☐ Other
- ☐ Riverbank Collapse
- ☐ Rock Fall
- ☐ Snow Avalanche

Downpour

Deadliest Trigger

Downpour

Most Common Cause

2.82

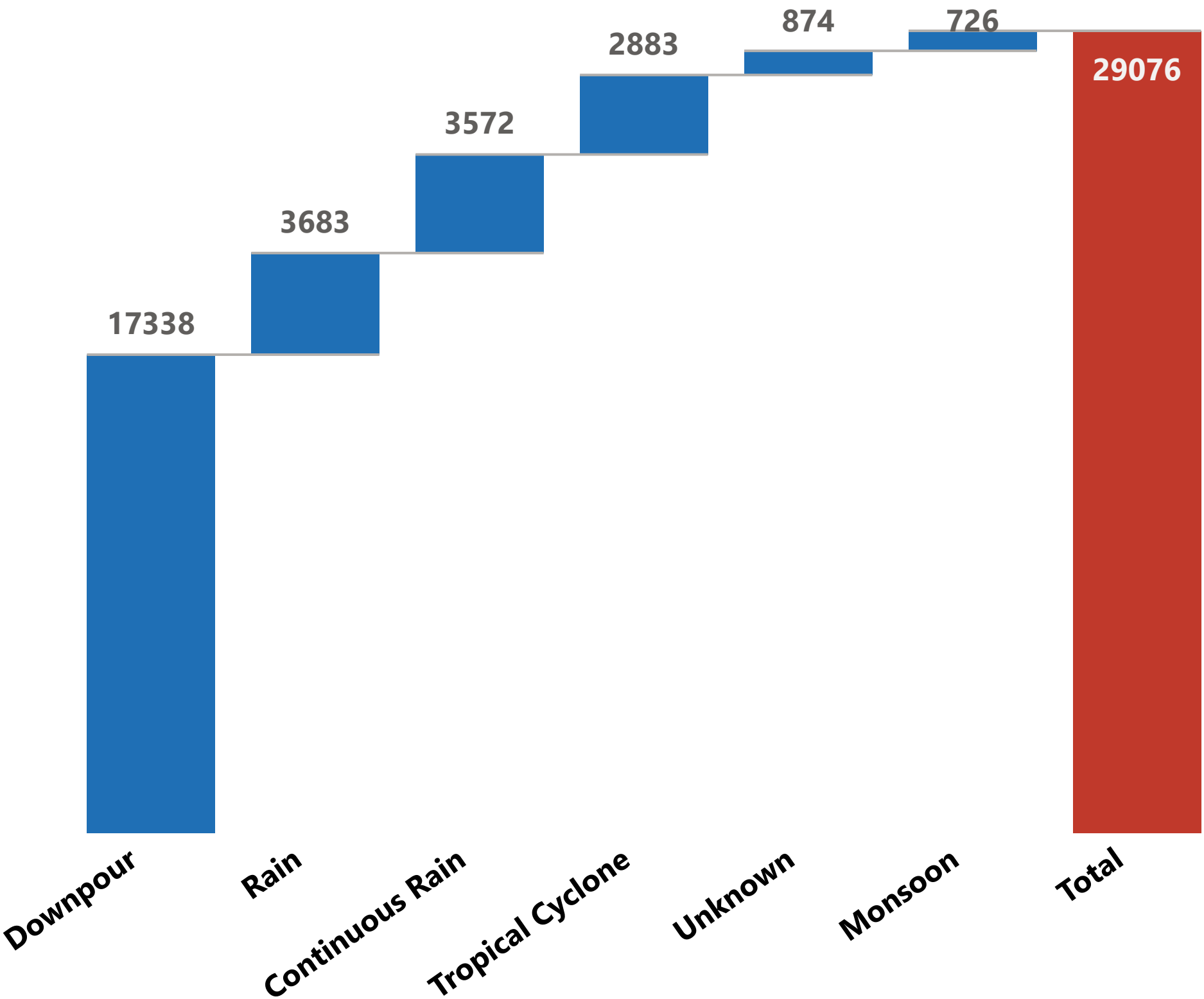
Avg Fatalities per Event

33

Catastrophic Events

Fatality Buildup by Trigger

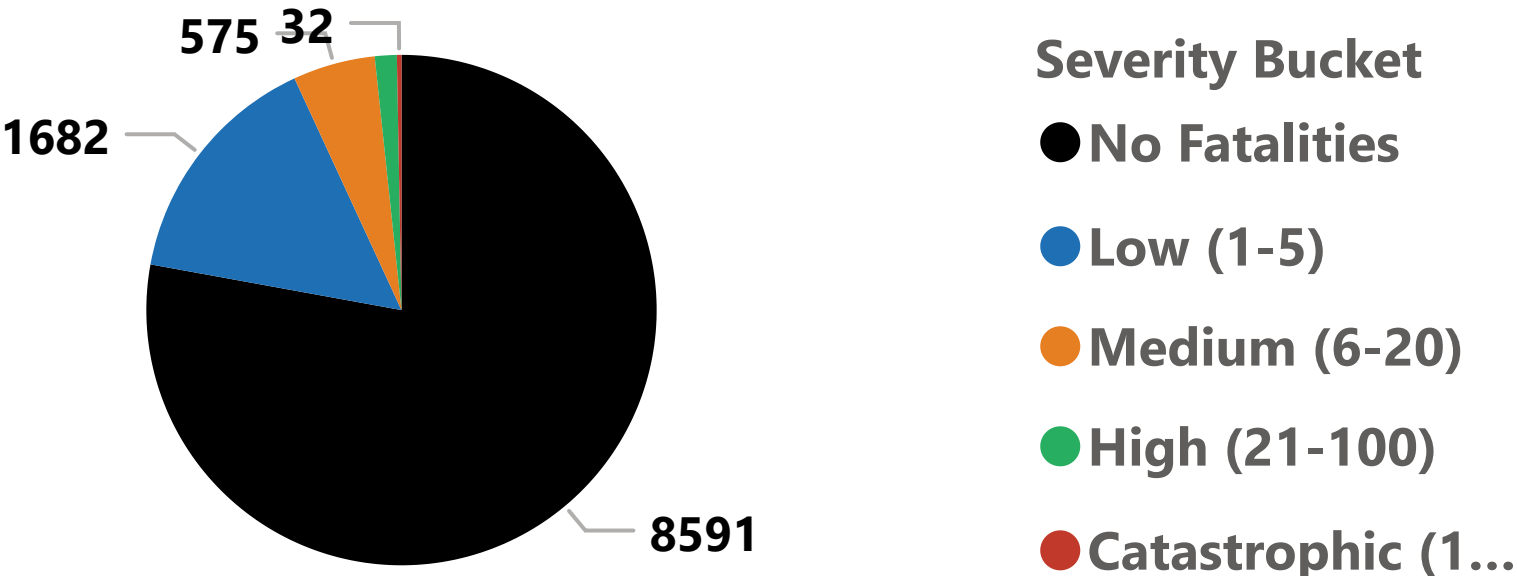
Contribution ● Increase ● Decrease ● Total



Trigger × Size — Event Count

Landslide Trigger	Catastrophic	Large	Medium	Small	Unknown
Construction		2	45	34	
Continuous Rain	1	50	423	255	
Dam Embankment Collapse		2	5	3	
Downpour	1	370	3280	944	
Earthquake		11	46	28	
Flooding		11	37	22	
Freeze Thaw	1	2	15	22	
Leaking Pipe				10	
Mining		6	56	28	
Total	3	750	6551	2767	

Severity Distribution





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Year

All

Country Name

All

Country Name

Country Name	Total Fatalities
India	7,069.00
China	4,945.00
Afghanistan	2,294.00
Philippines	1,847.00
Brazil	1,743.00
Indonesia	1,697.00
Nepal	1,473.00
Guatemala	743.00
Pakistan	662.00
Colombia	591.00
Taiwan	540.00
Uganda	534.00
Total	28,878.00

July

Peak Month

2010

Peak Year

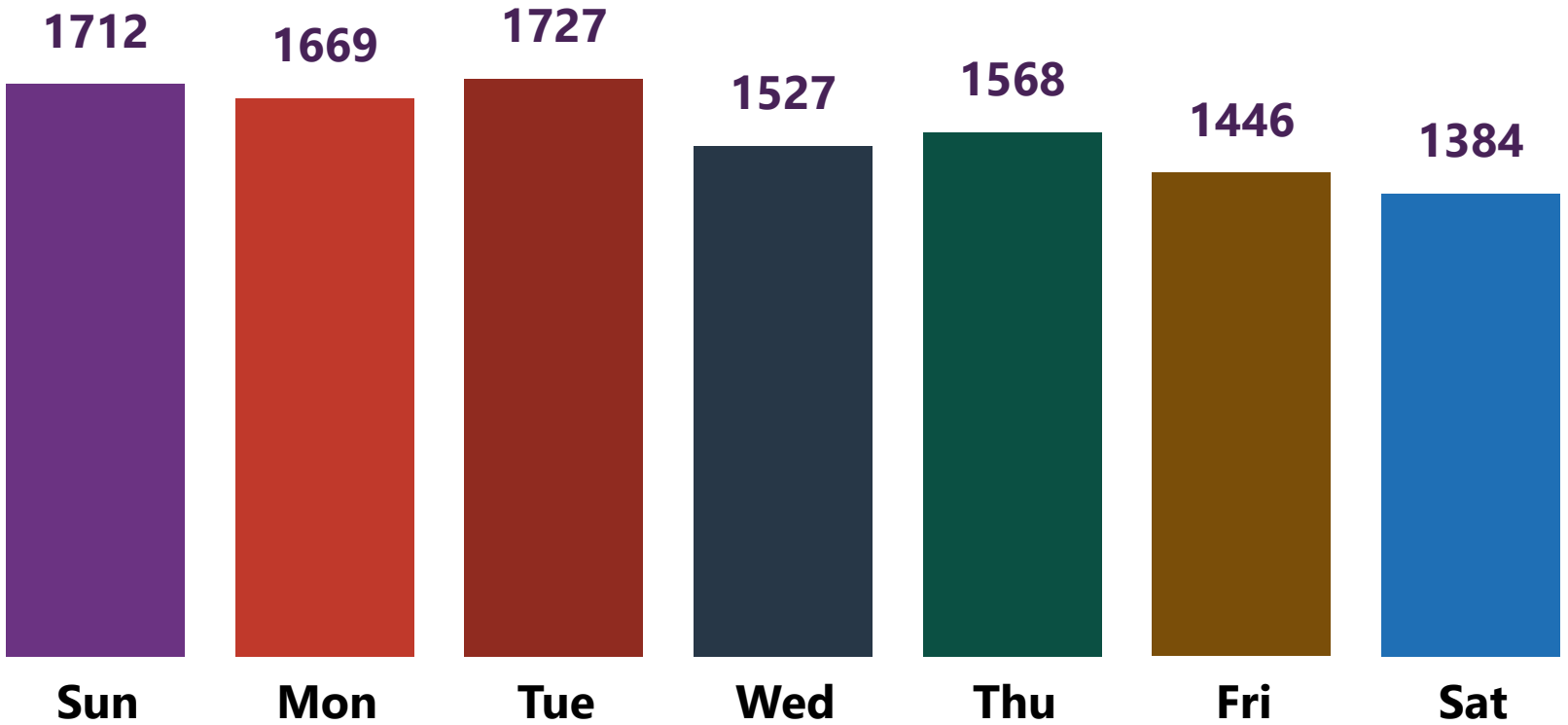
525.38

Avg Events per Year

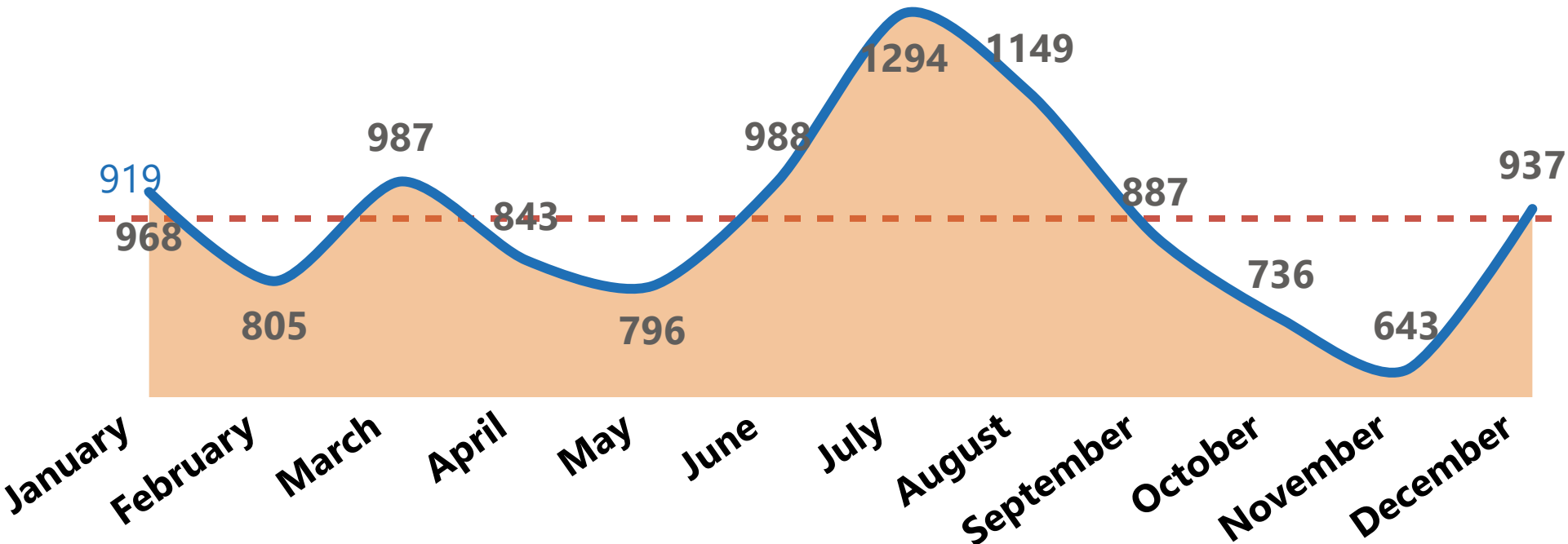
United States

Most Affected Country

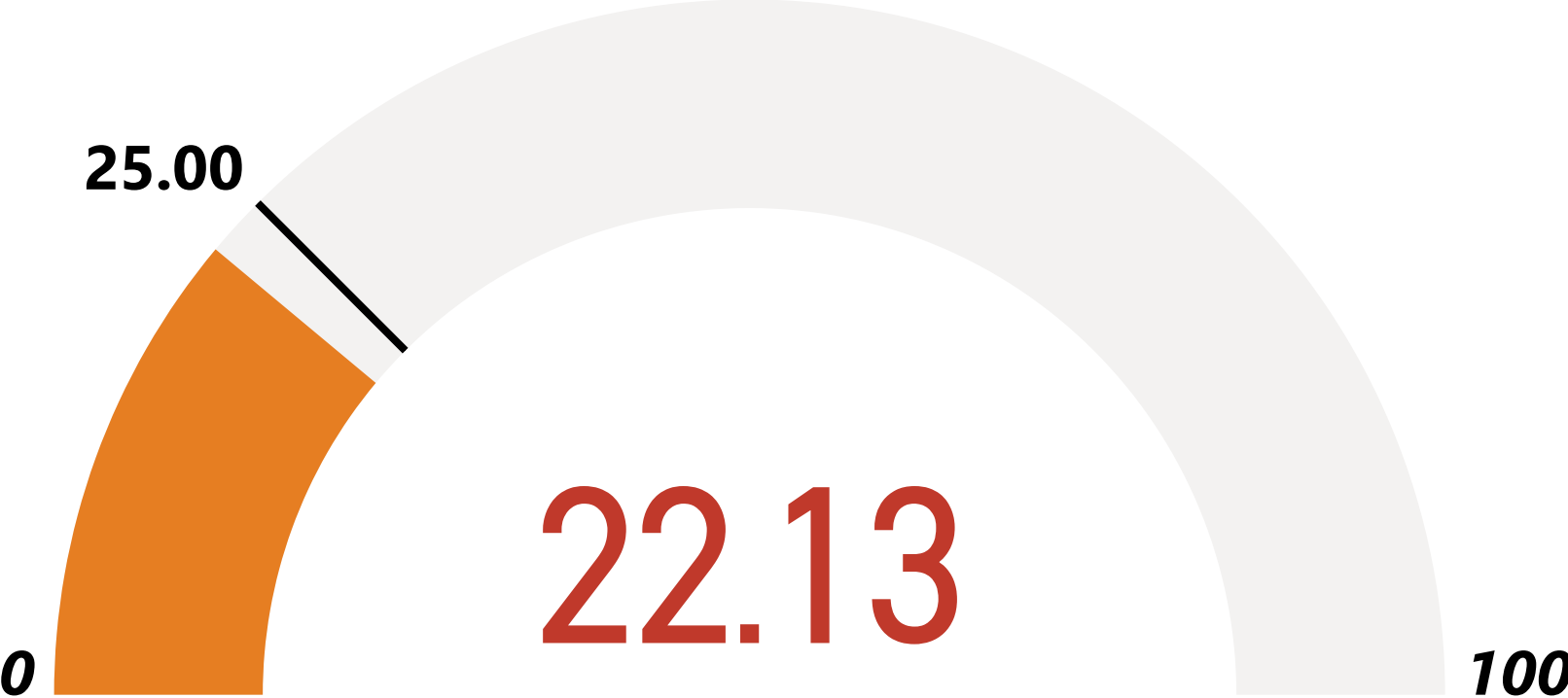
Events By Day Of Week



Total Events By Month Name



Fatality Rate — % of Events with Deaths



Total Events by Quarter

